Product Bulletin

TMS320DM310 DSP

High-Performance Digital Media Solution for the Portable Appliance Market

Key Features

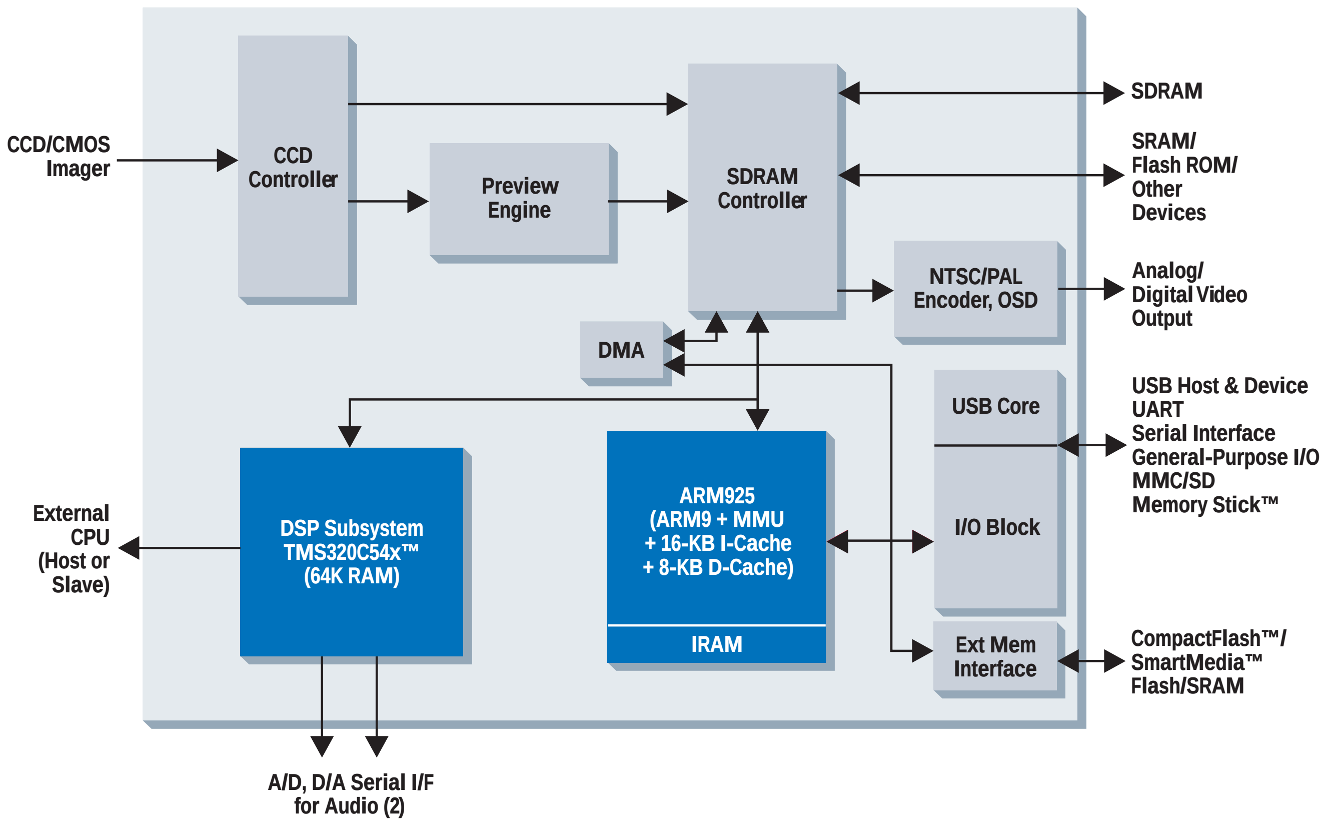
- Real-time MPEG-1, -4 video encode at CIF resolution (352 × 288)
- Real-time MPEG-4 video decode at VGA resolution (640 × 480)
- One-second shot-to-shot delay for 6-megapixel image
- Supports multiple applications and file formats, including AAC, H.263, H.264, JPEG, JPEG2K, M-JPEG, MP3, MPEG-4, MPEG-2, MPEG-1, WMA and WMV
- Highly integrated system-on-a-chip (SoC) design reduces overall system cost

New TI Technology for Portable Digital Media Applications

The new TMS320DM310 programmable DSP-based solution from Texas Instruments (TI) is a highly integrated multimedia engine offering unparalleled performance, leading-edge process technology and flexibility for next-generation portable audio-visual products. As one of the first devices of the new digital media family, the DM310 solution delivers more than three times the performance of TI's successful TMS320DSC2x processors. The flexible design enables the support of CMOS and CCD camera imagers up to 6 megapixels with less than a 1-second shot-to-shot delay. The system-on-a-chip (SoC) enables a new level of capabilities in a variety of digital media applications, including digital cameras, PC cameras, streaming media systems, digital video recorders (DVRs), personal video recorders (PVRs) and audio/video jukeboxes.

The DM310 processor is based on the same hardware platform

TMS320DM310 Functional Block Diagram



used in TI's successful DSC2x processor family. Hardware accelerators allow for real-time data processing of computationally intensive video and imaging algorithms. By using the industry-leading, powerful and power-efficient TMS320C5000™ DSP core along with powerful hardware accelerators (including 8 MAC units), the new device can achieve real-time MPEG-1 and MPEG-4 video encode at CIF resolution (352 × 288) and real-time MPEG-4 video decode at VGA resolution (640 × 480). This leaves the powerful ARM9 microprocessor with cache to drive other system functions such as user interface or simultaneous audio Codecs.

Leading-edge 0.13-micron process technology allows for a small product footprint and power-efficient operation. In addition, state-of-the-art power-down and standby modes decrease overall power consumption for longer system use by consumers.

The DM310 processor improves the feature set of the previous generations by adding USB host, MultiMediaCard (MMC), Secure Digital™ and Memory Stick™ interfaces. The high-level integration in the DM310 processor saves board space and lowers overall system cost. This high-level performance enables new combinations of video, imaging, audio and voice in portable consumer media applications.

Extensive software, development tools and third-party support for the DM310 DSP help to simplify system development and speed time to market.

ARM System Controller

The DM310 ARM subsystem includes an ARM9 32-bit RISC processor with 16-KB instruction cache, 8-KB data cache and a robust memory management unit. The ARM925 CPU performs general system control such as system initialization, configuration and graphical user interface (GUI) or can be used for additional general-purpose processing.

DSP Subsystem

The DM310 DSP subsystem consists of the industry-proven, power-efficient TMS320C5000 DSP and performance-enhancing programmable hardware accelerators. The DSP subsystem provides enough processing power to handle various industry-standard, computationally intensive imaging, audio and video algorithms. The DSP subsystem is fully programmable, allowing the designer to implement his own image processing algorithms and to provide functionality upgrades even after the product is in the hands of consumers.

Comprehensive Support Speeds Time to Market

Getting Started

The DM310 evaluation module (EVM) is a fully functioning camera designed to make prototyping, development and debugging faster, which speeds time to market. The DM310 DSP's object code compatibility with the DSC2x generation allows developers to begin programming using existing DSC platforms, then port the development software to newer devices in the family as they become available.

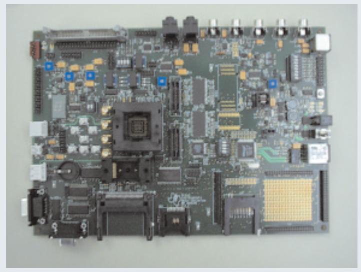
Technical Details

- Industry-recognized, power-efficient TMS320C54x™ DSP core with 64K program and data RAM
- Versatile ARM9 32-bit RISC processor with 16-KB I-cache and 8-KB D-cache
- Hardware accelerators speed video-imaging processing, encoding and decoding
- State-of-the-art 0.13-micron (GS40) CMOS process technology
- Design is code-compatible with the DSC2x platform
- Support for up to 16-MB Flash memory
- Universal serial bus (USB) host and device support
- Five timers (four general-purpose, one watchdog)
- Integrates one UART and two serial interfaces
- Intelligent power management
- Preview engine for real-time live-view function
- Versatile on-screen display (OSD) support with multiple windows, zoom, double buffering, blending and scaling
- Flexible SDRAM controller capable of up to 396-MB SDRAM bandwidth; supports single 16-bit SDRAM, or multiple 16-/32-bit SDRAM devices
- Real-time MPEG-1, -4 video encode at CIF resolution (352 × 288)
- Real-time MPEG-4 video decode at VGA resolution (640 × 480)
- Ability to run networking protocols including H.323/324
- Seamless interface to MMC, Secure Digital™, Memory Stick™, CompactFlash™ and SmartMedia™
- Programmable support for MP3/Dolby℠ AC3/G.7xx audio decode and playback
- Programmable support for auto focus, auto exposure and auto white balance
- Support for multiple operating systems including Nucleus™, Linux™, uITRON, VxWorks™ and WinCE
- Digital LCD interface
- NTSC/PAL/RGB video output
- 288 GHK MicroStar BGA™ package (16 × 16 × 0.8 mm)

Code Composer Studio™ Development Tools

Code Composer Studio™ is a fully integrated development environment (IDE) supporting TI's DM310 digital media processor. It is one of the key components of eXpressDSP™ Software and Development Tools that slashes development and integration time for DSP software. Its complete and easy-to-use set of development tools and features addresses each phase of the code development cycle, including editing, building, debugging, code profiling and project management.

DM310 EVM Features



- TMS320DM310 DSP
- 2-MB Flash/512-KB SRAM
- 16-MB SDRAM
- Single 5-V power supply
- Power management and built-in Li-Ion battery charger
- JTAG-based ARM and DSP emulation support
- CCD/CMOS connector and power-supply support
- CompactFlash™, SmartMedia™, Secure Digital™, MMC and Memory Stick™ storage media support
- USB device and USB host communication support
- UART communication support
- NTSC/PAL composite and S-video and analog RGB outputs
- Digital LCD interface
- Audio Codec (TMS320AIC23) with microphone/headphone support
- GIO push-button inputs and LED outputs
- Expansion connector
- Ethernet port

DSP/BIOS™ Kernel II

This scalable, extensible, real-time kernel for the DM310 DSP subsystem provides a standard software foundation to significantly reduce cost, risk and development time.

Third-Party Network

To facilitate in providing a total system solution, TI is leveraging and expanding its extensive Third-Party Network. These network members have developed DM310 processor solutions for video, imaging and audio algorithm development, system integration and operating system support. Their proven performance enables customers to significantly reduce the time elapsed from concept to market to revenue.

For More Information

To obtain more information on how the DM310 DSP can enhance your imaging system, please contact your local TI field sales office or TI Product Information Center, or visit the TI imaging Web site at:

www.ti.com/dm310

TI Worldwide Technical Support

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support.ti.com

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